EBIKE101 Day 1

Introduction, Rider Safety, and System Inspection

Safe micro-mobility starts before the throttle moves.

Today's question

What makes an e-bike different from a bicycle, and what must be proven safe before anybody rides it?

Day 33 | Monday, August 31, 2026

Core rule

No ride.
No throttle test.
No powered wheel spin.

Until the bike passes a visible safety check.

A bicycle is already a system

Frame, fork, wheels, tires, brakes, drivetrain, cockpit, and rider contact points.

Electricity does not fix loose wheels.

An e-bike adds more systems

Battery, controller, motor, display, throttle or pedal assist, wiring, connectors, charger, and cutoffs.

More capability means more inspection. Amazing how physics keeps having opinions.

Tires. Brakes. Wheels. Steering. Drivetrain. Seat. Pedals. Frame and fork.

If the bicycle is unsafe, the e-bike is unsafe.

Battery condition. Battery mount. Controller mount. Wiring path. Connectors. Display. Throttle or PAS. Charger.

Turning on is not the same as safe.

Rider input -> controller -> battery power -> motor -> wheel motion

The rider asks. The controller decides. The motor moves.

Battery boundary

No swollen, hot, damaged, wet, unknown, or mismatched battery or charger gets used.

Stored energy deserves respect.

Wiring boundary

No wires near tires, chains, cranks, brake rotors, sharp edges, heat, or pinch points.

Loose wires are future problems volunteering early.

Pre-ride categories

Mechanical. Electrical. Rider control. Area and route. Documentation.

Pass, fail, or instructor review. No fantasy category called “probably fine.”

Stop-work triggers

Unexpected motion. Loose battery. Failed brakes. Wheel rub. Damaged wiring. Unknown charger. Unsafe joke becoming unsafe behavior.

Pause the work before the work teaches the lesson the hard way.

Inspector 1: mechanical
Inspector 2: electrical
Recorder: notes
Reporter: debrief

Roles keep the lab from becoming six humans orbiting one bike.

Lab A

Inspect the mechanical bike first.

Frame, fork, wheels, tires, brakes, steering, drivetrain, contact points.

Lab B

Inspect the e-bike systems second.

Battery, motor, controller, wiring, connectors, display, throttle/PAS, charger, cutoffs.

Lab C

Draw the system flow and mark what could fail.

Input -> control -> power -> motion -> rider.

Winning condition

Find it. Name it. Document it. Explain the safe next step.

No guessing. No sending it. Humanity has tested that.

Debrief questions

What mechanical risk did you find?
What electrical risk did you find?
What rider-control risk did you find?

What must be checked first tomorrow?

Tomorrow

Bicycle mechanics foundations.

Before conversion, the bike itself has to be worthy of electricity.

Quote of the day

Power only comes after control is proven.

EBIKE101 | CTS Cohort 6